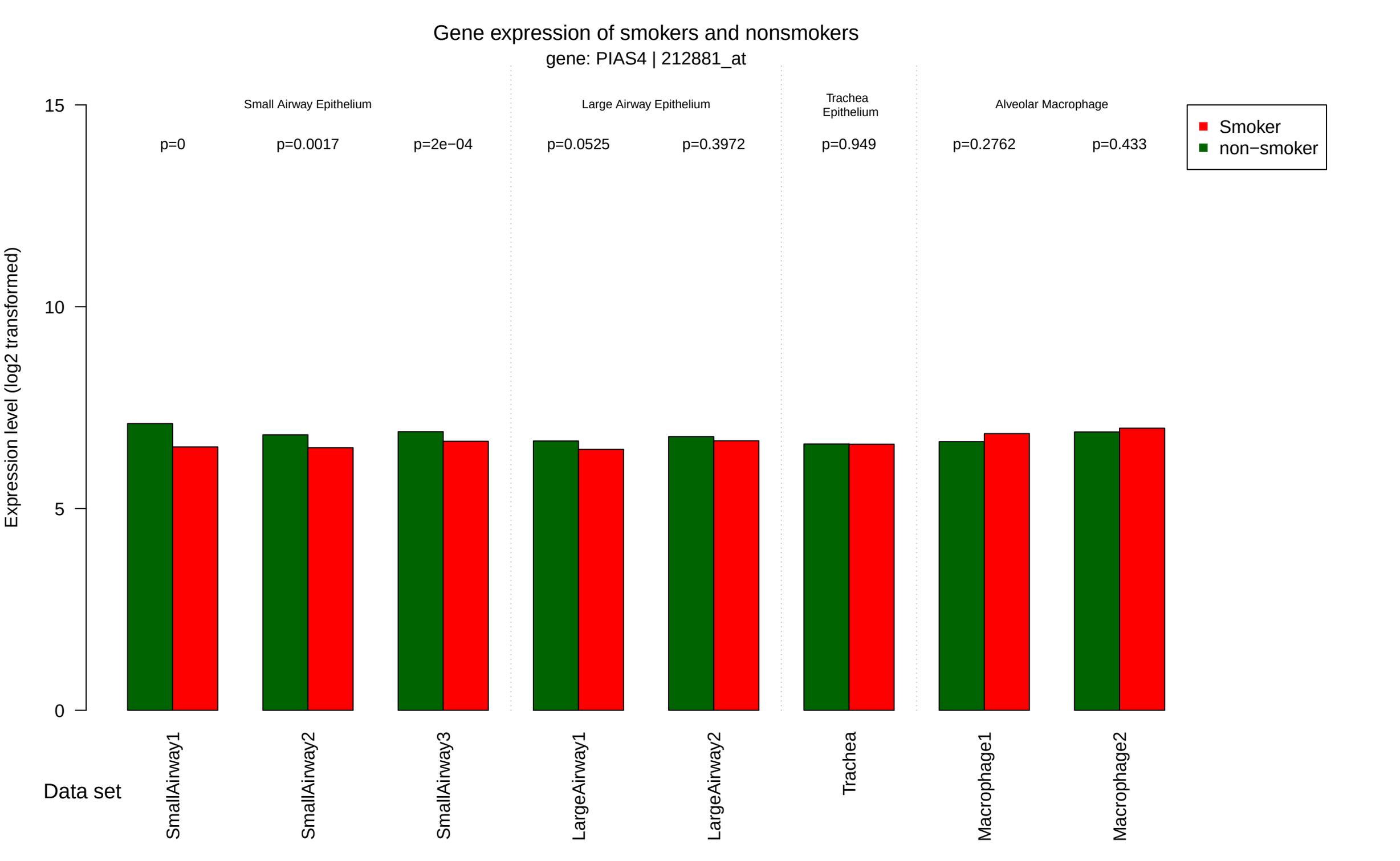
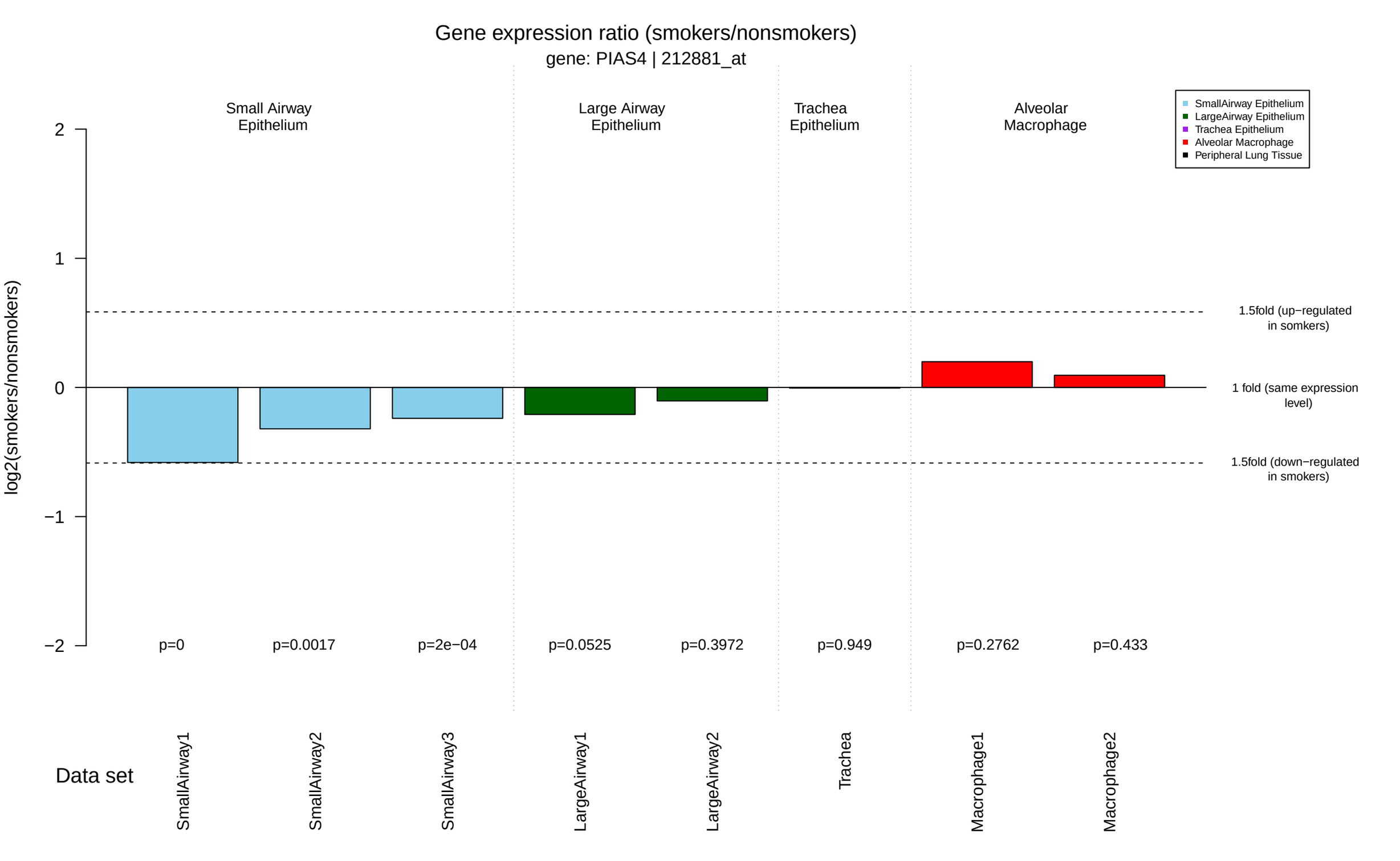
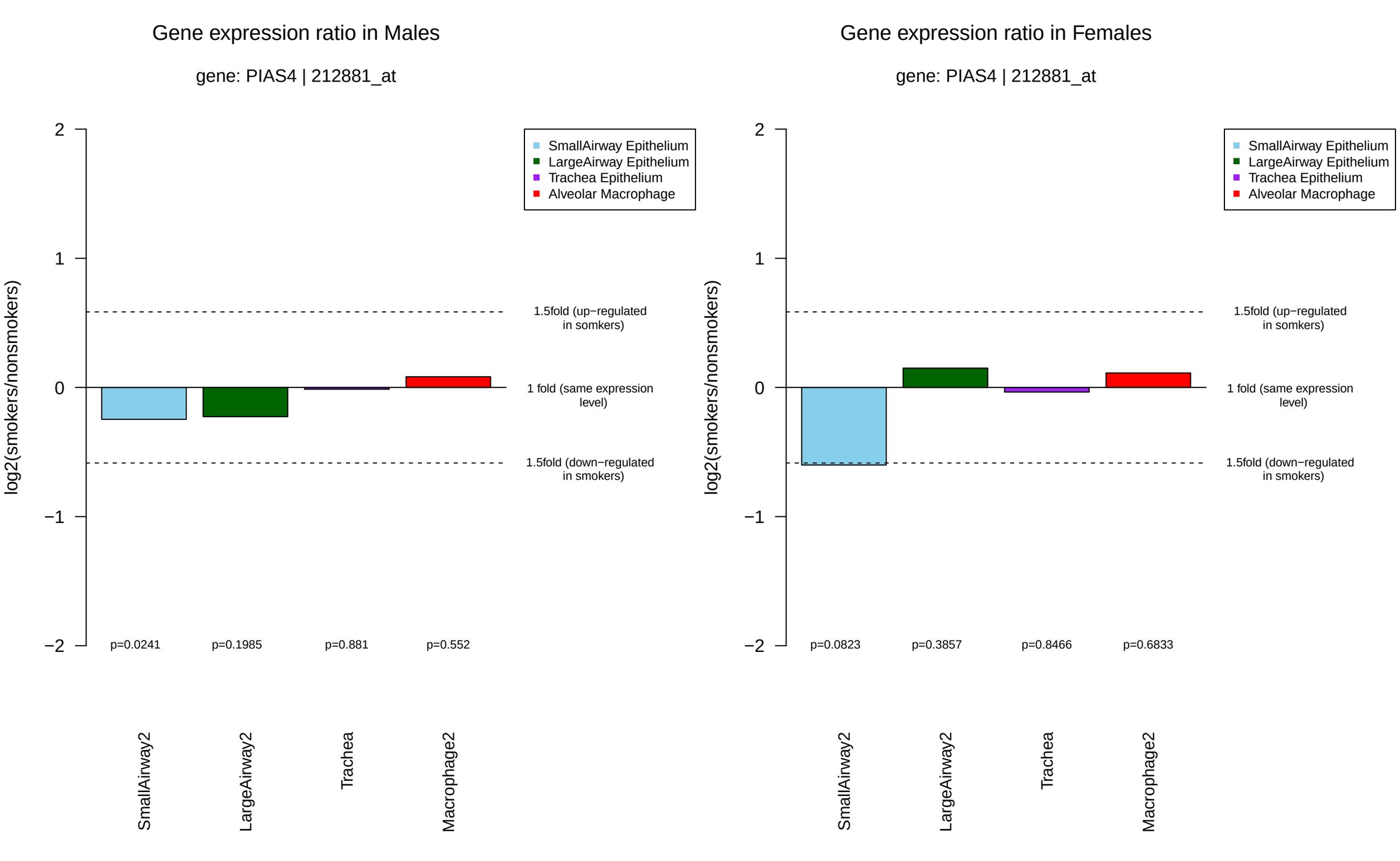
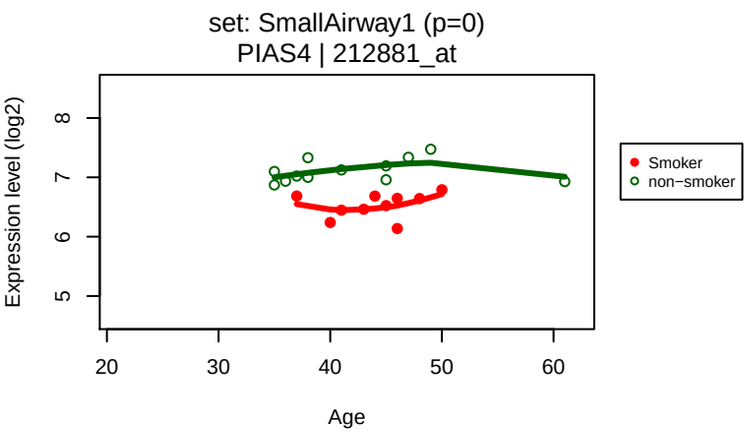
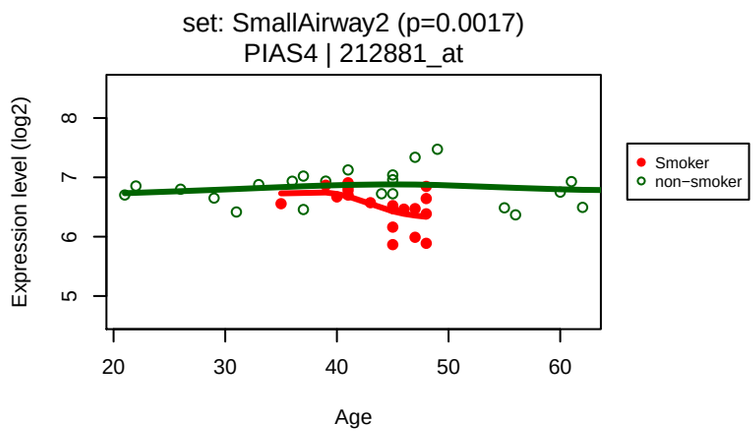
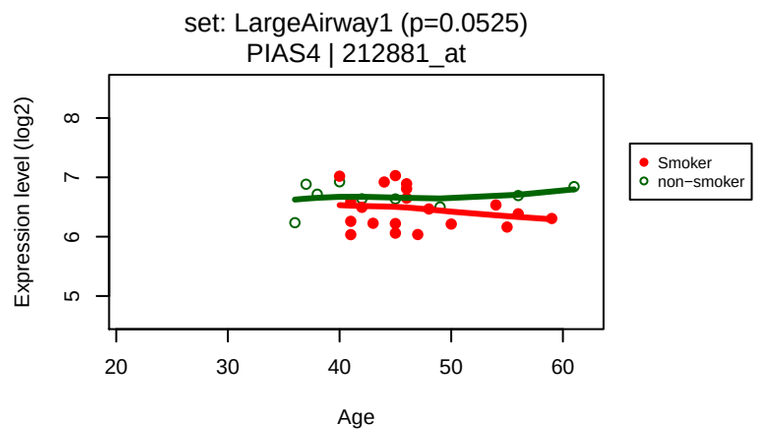

Gene expression of smokers and nonsmokers
gene: PIAS4 | 212881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0
p=0.0017
p=2e−04
p=0.0525
p=0.3972
p=0.949
p=0.2762
p=0.433
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

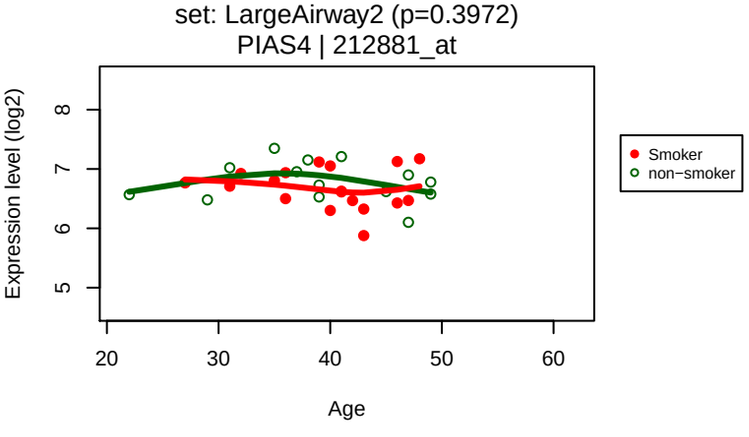

Gene expression ratio (smokers/nonsmokers)
gene: PIAS4 | 212881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0
p=0.0017
p=2e−04
p=0.0525
p=0.3972
p=0.949
p=0.2762
p=0.433
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

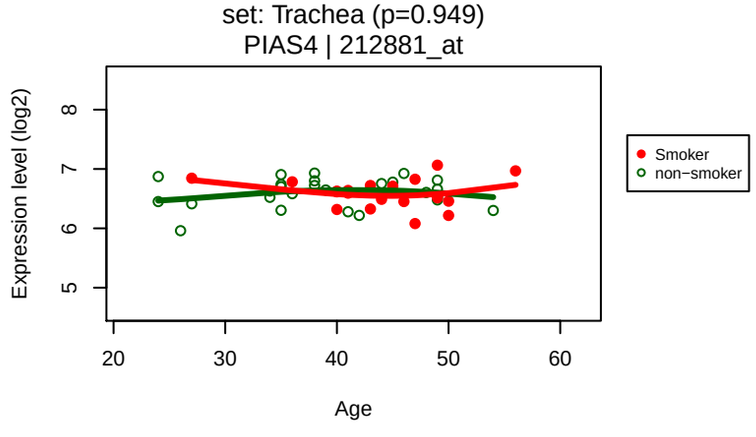

Gene expression ratio in Males
gene: PIAS4 | 212881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0241
p=0.1985
p=0.881
p=0.552
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS4 | 212881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0823
p=0.3857
p=0.8466
p=0.6833
SmallAirway2
LargeAirway2
Trachea
Macrophage2

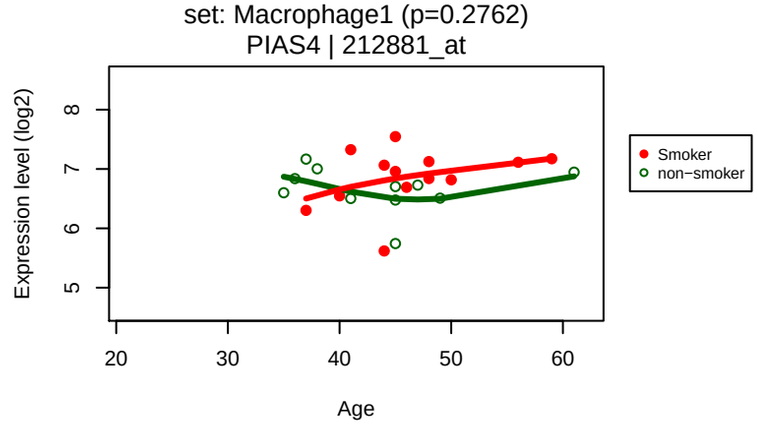

set: SmallAirway1 (p=0)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker

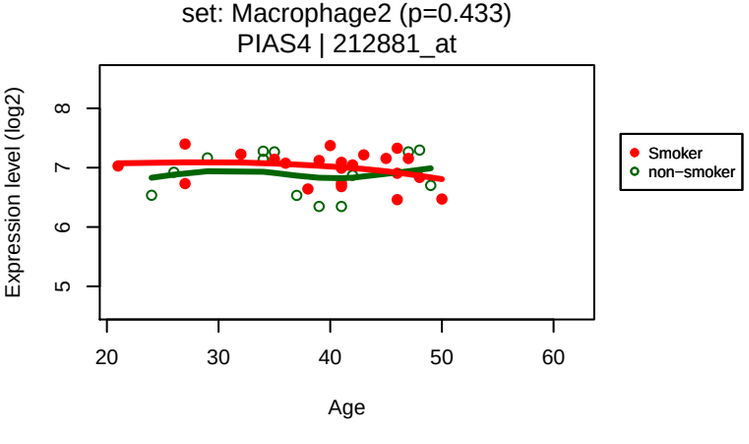

set: SmallAirway2 (p=0.0017)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker

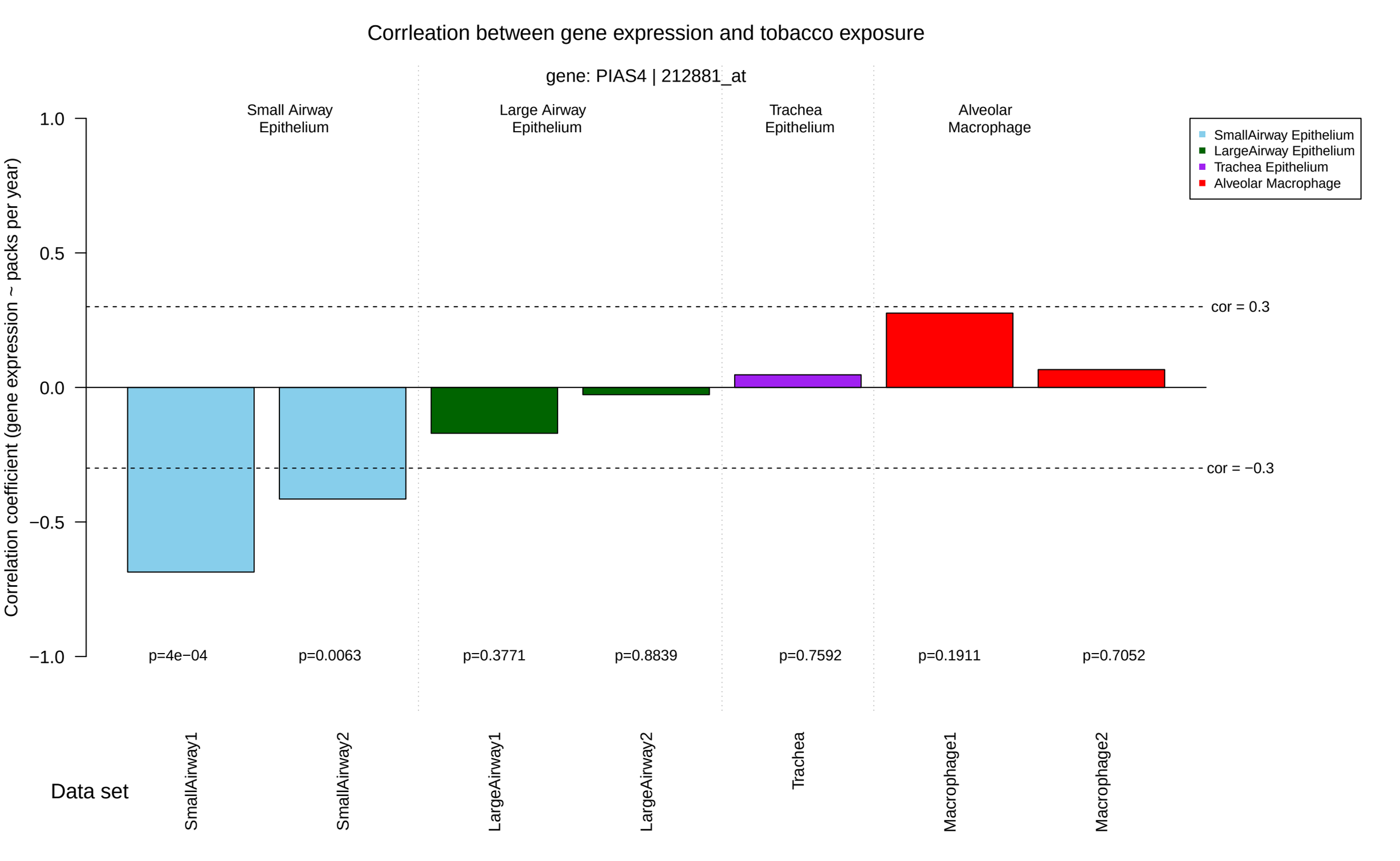

set: LargeAirway1 (p=0.0525)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3972)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.949)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2762)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.433)
PIAS4 | 212881_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS4 | 212881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=4e−04
p=0.0063
p=0.3771
p=0.8839
p=0.7592
p=0.1911
p=0.7052
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2